

Stat 5200, Module 4: Factorial Design and Analysis

Jesse Wheeler

Contents

1	Overview	1
2	Factorial Treatment Structure	1
3	Understanding Interaction Effects	2
4	The Two-Way ANOVA Model	3
5	Factorial Contrasts	4

1 Overview

Objectives and overview

- This material is based on Chapter 8 of the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Module outcomes include:
 - You will understand complete and incomplete factorial treatment structures.
 - You will develop an intuitive and formal understanding of *interaction effects*.
 - You will be able to specify and fit additive and non-additive two-way ANOVA models.
 - You will learn how to construct and interpret main effect contrasts and interaction contrasts.
 - You will master the model testing sequence (testing interaction first).
 - You will know how to fit and analyze factorial designs in R using `aov()` and `emmeans()`.

2 Factorial Treatment Structure

Introduction to Factorial Structures

- In many experiments, we want to evaluate the effect of *two or more explanatory factors* simultaneously.
- A *factorial treatment structure* defines treatments as combinations of levels of each factor.
- Suppose Factor A has a levels and Factor B has b levels.
- A *complete factorial design* includes all $g = a \times b$ possible treatment combinations.

Example 1: 2×3 Diet Experiment

- Factor 1: Protein Source (Beef, Cereal, Pork) $\implies b = 3$.
- Factor 2: Protein Intake (High, Low) $\implies a = 2$.
- Total treatment combinations: $g = 2 \times 3 = 6$ unique diets.

Example 2: 2×2 Fertilizer Experiment

- Factor A: Superphosphate (P) $\rightarrow$ Levels: {Not Added, Added}
- Factor B: Potash (K) $\rightarrow$ Levels: {Not Added, Added}
- $g = 2 \times 2 = 4$ treatments:
 1. None (P_1K_1)
 2. Superphosphate only (P_2K_1)
 3. Potash only (P_1K_2)
 4. Both Superphosphate and Potash (P_2K_2)

Notice the power of a factorial design: instead of running two separate experiments (one for Superphosphate and one for Potash), we run a single experiment that evaluates both factors simultaneously using the exact same experimental units. This yields higher statistical efficiency and allows us to study whether the two factors work together or interfere with each other.

Notation for Two-Way Factorials

- To identify observations in a two-way design, we use three subscripts:

$$y_{ijk}$$

- $i = 1, \dots, a$: level of Factor A
- $j = 1, \dots, b$: level of Factor B
- $k = 1, \dots, n$: replication number within treatment combination (i, j)

- Summary Notation:

- Sample mean for treatment (i, j) : $\bar{y}_{ij\bullet} = \frac{1}{n} \sum_{k=1}^n y_{ijk}$
- Marginal mean for level i of Factor A: $\bar{y}_{i\bullet\bullet} = \frac{1}{bn} \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$
- Marginal mean for level j of Factor B: $\bar{y}_{\bullet j\bullet} = \frac{1}{an} \sum_{i=1}^a \sum_{k=1}^n y_{ijk}$
- Overall grand mean: $\bar{y}_{\bullet\bullet\bullet} = \frac{1}{abn} \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$

3 Understanding Interaction Effects

What is an Interaction Effect?

- *Definition*: An interaction effect occurs when the effect of one factor on the response depends on the level of another factor.
- Ask yourself: "Does the difference between levels of Factor A stay the same regardless of Factor B?"
 - If **YES**: The factors are *additive* (No Interaction).

- If **NO**: There is an *interaction* between factors.

Intuition: "Difference of Differences"

Consider a 2×2 design with population means μ_{ij} :

- Simple main effect of A when Factor B is at level 1: $\mu_{11} - \mu_{21}$
- Simple main effect of A when Factor B is at level 2: $\mu_{12} - \mu_{22}$

The *interaction effect* is the difference between these two simple effects:

$$(\mu_{11} - \mu_{21}) - (\mu_{12} - \mu_{22})$$

If this difference equals zero, the effect of A is constant across levels of B. If non-zero, an interaction exists.

Graphical Diagnosis: Interaction Plots

- An *interaction plot* displays treatment group means across levels of one factor, with lines connecting levels of the second factor.
- **Parallel lines** $\implies$ No Interaction (Additive).
- **Non-parallel or crossing lines** $\implies$ Interaction present.

4 The Two-Way ANOVA Model

Model Specification

- We model the response y_{ijk} as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- μ : Overall grand mean.
- α_i : Main effect of level i of Factor A.
- β_j : Main effect of level j of Factor B.
- $(\alpha\beta)_{ij}$: Interaction effect between level i of A and level j of B.
- $\epsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2)$: Random errors.
- *Parameter Identification Restrictions (Sum-to-Zero Constraints)*:

$$\sum_{i=1}^a \alpha_i = 0, \quad \sum_{j=1}^b \beta_j = 0, \quad \sum_{i=1}^a (\alpha\beta)_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^b (\alpha\beta)_{ij} = 0 \text{ for all } i$$
- *Hierarchy of Models*:
 1. **Full (Non-additive) Model**: $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$
 2. **Additive Model**: $y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$
 3. **Main Effect A Model**: $y_{ijk} = \mu + \alpha_i + \epsilon_{ijk}$
 4. **Constant Model**: $y_{ijk} = \mu + \epsilon_{ijk}$

5 Factorial Contrasts

Main Effects vs. Interaction Contrasts

- A contrast in a factorial design is a linear combination of treatment means $\sum_i \sum_j w_{ij} \mu_{ij}$ subject to $\sum_i \sum_j w_{ij} = 0$.

- *Main Effect Contrasts for A*: Weights for a given level of A are identical across all levels of B.

$$\sum_{i=1}^a w_i = 0, \quad w_{ij} = w_i \quad \text{for all } j$$

- *Interaction Contrasts ("Contrast of Contrasts")*: Weights are constructed as products of main effect contrast weights: $w_{ij} = w_{A,i} \times w_{B,j}$. Row sums and column sums of weights must **all equal zero**:

$$\sum_{i=1}^a w_{ij} = 0 \text{ for all } j, \quad \text{and} \quad \sum_{j=1}^b w_{ij} = 0 \text{ for all } i$$

Example: 2×2 Factorial Weight Matrices

Main Effect A (W_A)

1	1
-1	-1

Main Effect B (W_B)

1	-1
1	-1

Interaction (W_{AB})

1	-1
-1	1

```
# Create factorial dataset
fert_data <- data.frame(
  P = factor(rep(c("No_P", "P"), each = 8)),
  K = factor(rep(rep(c("No_K", "K"), each = 4), 2)),
  yield = c(183, 176, 291, 254, # None
            224, 258, 244, 214, # K only
            356, 300, 301, 271, # P only
            329, 283, 308, 326) # P and K
)
```

```
# Display sample means across cell combinations
with(fert_data, tapply(yield, list(P, K), mean))
```

```
# Generate interaction plot
interaction.plot(x.factor = fert_data$P, trace.factor = fert_data$K,
  response = fert_data$yield, fun = mean,
  type = "b", pch = c(16, 17), col = c("darkred", "darkblue"),
  xlab = "Superphosphate (P)", ylab = "Mean Yield",
  trace.label = "Potash (K)")
```

```
# Fit Two-Way ANOVA Model with Interaction
fert_mod <- aov(yield ~ P * K, data = fert_data)
summary(fert_mod)
```

Acknowledgments

- Compiled on October 1, 2026 using R version 4.5.0.
- Licensed under the [Creative Commons Attribution-NonCommercial license](#).  Please share and remix non-commercially, mentioning its origin.
- We acknowledge [students and instructors for previous versions of this course / slides](#).

References

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.